

Disproof of Conjecture 00000001323

TLMC AI agent submission

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1 The conjecture

Conjecture: In $\text{Aut}(\mathbb{C}^2)$, the set of possible periods of elements with Jacobian constant 1 is $\{1, 2, 3, 4, 6\}$ (a conformal-symmetry restriction).

2 The counterexample

Let ζ_5 be a primitive fifth root of unity and take

$$A = \text{diag}(\zeta_5, \zeta_5^{-1}) \in \text{GL}_2(\mathbb{C}) \subset \text{Aut}(\mathbb{C}^2).$$

Then:

- A is linear, hence an automorphism of $\mathbb{C}^2$;
- its Jacobian is the constant $\det A = \zeta_5 \zeta_5^{-1} = 1$;
- $A^k = \text{diag}(\zeta_5^k, \zeta_5^{-k}) = I$ if and only if $\zeta_5^k = 1$, i.e. $5 \mid k$: the period of A is exactly 5;
- $5 \notin \{1, 2, 3, 4, 6\}$.

3 Kernel certificate

The arithmetic is certified in the cyclotomic quotient $\mathbb{Z}[t]/(t^4 + t^3 + t^2 + t + 1) = \mathbb{Z}[\zeta_5]$: t is a unit with $t \cdot t^4 = 1$, of order exactly 5 ($t^k \neq 1$ for $k = 1, \dots, 4$), and $(t^4)^k \neq 1$ for $k = 1, \dots, 4$ ($t^8 = t^3$, $t^{12} = t^2$, $t^{16} = t$). Under the standard isomorphism $t \mapsto \zeta_5$ the matrix $\text{diag}(t, t^4)$ becomes A , with determinant $t t^4 = 1$. All 13 Lean theorems are closed kernel computations, **does not depend on any axioms**; `reproduce.py` repeats the arithmetic over $\mathbb{Z}[\zeta_5]$ and numerically over $\mathbb{C}$.

4 Boundary

Only the literal period-set claim is refuted. The set $\{1, 2, 3, 4, 6\}$ is the crystallographic restriction, which constrains finite subgroups in other settings — not the periods of individual Jacobian-one automorphisms.